

ImPRESS Model selection

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Introduction

In this document we want to assess the model fit for different covariance structures in the primary statistical model used in ImPRESS. We use the Akaike Information Criterion (AIC) as the model selection criteria. The four models we want to compare are: unstructured, autoregressive and compound symmetry. Note that these analyses are based on sham randomisation, we deliberately ignore the true treatment allocation to avoid any information bias.

Analyses

```
adeff <- tar_read(adeff, store = "_targets")
tmp <- adeff |>
  filter(paramcd == "mrt1cf" & cohortcd == 2) |>
  filter(avisitn <= 43) |>
  mutate(trt = case_when(
    avisitn == 15 & str_starts(step, "Cycle 1") ~ paste0(dose01p, " ", dose01pu),
    avisitn == 29 & str_starts(step, "Cycle 1") ~ paste0(dose01p, " ", dose01pu),
    avisitn == 43 & str_starts(step, "Cycle 1") ~ paste0(dose01p, " ", dose01pu),
    avisitn == 29 & str_starts(step, "Cycle 2") ~ paste0(dose01p, " ", dose01pu),
    avisitn == 43 & str_starts(step, "Cycle 2") ~ paste0(dose01p, " ", dose01pu),
    avisitn == 43 & str_starts(step, "Cycle 3") ~ paste0(dose01p, " ", dose01pu),
    TRUE ~ "0mg"
  )) |>
  mutate(
    trt = factor(trt, levels = c("0mg", "25 mg", "50 mg", "100 mg")),
    subjid = factor(subjid),
    avisitn = factor(avisitn, levels = c(-7, 15, 29, 43))
  )

m <- mmrm(aval ~ trt + avisitn + us(avisitn | subjid),
  data = tmp
)

m2 <- mmrm(aval ~ trt + avisitn + ar1(avisitn | subjid),
  data = tmp
)
```

```

m3 <- mmrm(aval ~ trt + avisitn + cs(avisitn | subjid),
  data = tmp
)
summary(m)

## mmrm fit
##
## Formula:      aval ~ trt + avisitn + us(avisitn | subjid)
## Data:         tmp (used 189 observations from 54 subjects with maximum 4
## timepoints)
## Covariance:   unstructured (10 variance parameters)
## Method:       Satterthwaite
## Vcov Method:  Asymptotic
## Inference:    REML
##
## Model selection criteria:
##      AIC      BIC    logLik deviance
##    106.1    126.0   -43.1     86.1
##
## Coefficients:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   1.48232    0.05661  53.00000   26.185  <2e-16 ***
## trt25 mg       0.01277    0.06659 110.49000    0.192   0.8483
## trt50 mg      -0.08002    0.06867 112.20000   -1.165   0.2464
## trt100 mg     -0.06386    0.07178 111.09000   -0.890   0.3756
## avisitn15      0.08581    0.03823  55.68000    2.245   0.0288 *
## avisitn29      0.07977    0.05323  75.65000    1.499   0.1381
## avisitn43     -0.01214    0.06938 118.91000   -0.175   0.8614
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Covariance estimate:
##      -7      15      29      43
## -7 0.1731 0.1747 0.1759 0.1512
## 15 0.1747 0.2261 0.2080 0.1805
## 29 0.1759 0.2080 0.2551 0.1996
## 43 0.1512 0.1805 0.1996 0.2331

summary(m2)

```

```

## mmrm fit
##
## Formula:      aval ~ trt + avisitn + ar1(avisitn | subjid)
## Data:         tmp (used 189 observations from 54 subjects with maximum 4
## timepoints)
## Covariance:   auto-regressive order one (2 variance parameters)
## Method:       Satterthwaite
## Vcov Method:  Asymptotic
## Inference:    REML
##
## Model selection criteria:
##      AIC      BIC    logLik deviance
##    105.4    109.3   -50.7     101.4
##

```

```

## Coefficients:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  1.48232    0.06217  79.96000  23.843  <2e-16 ***
## trt25 mg      0.03696    0.06726 135.72000   0.549   0.5836
## trt50 mg     -0.05700    0.07015 138.92000  -0.813   0.4179
## trt100 mg    -0.07982    0.07393 142.56000  -1.080   0.2822
## avisitn15     0.08448    0.04136 125.19000   2.043   0.0432 *
## avisitn29     0.07287    0.05938 136.33000   1.227   0.2219
## avisitn43    -0.02598    0.07625 143.94000  -0.341   0.7338
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Covariance estimate:
##      -7      15      29      43
## -7 0.2087 0.1765 0.1492 0.1261
## 15 0.1765 0.2087 0.1765 0.1492
## 29 0.1492 0.1765 0.2087 0.1765
## 43 0.1261 0.1492 0.1765 0.2087
summary(m3)

## mmrm fit
##
## Formula:      aval ~ trt + avisitn + cs(avisitn | subjid)
## Data:         tmp (used 189 observations from 54 subjects with maximum 4
## timepoints)
## Covariance:   compound symmetry (2 variance parameters)
## Method:       Satterthwaite
## Vcov Method:  Asymptotic
## Inference:    REML
##
## Model selection criteria:
##      AIC      BIC    logLik deviance
##    104.3    108.3    -50.2    100.3
##
## Coefficients:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  1.48232    0.06367  69.24000  23.281  <2e-16 ***
## trt25 mg      0.02240    0.06931 141.82000   0.323   0.747
## trt50 mg     -0.04898    0.07110 142.71000  -0.689   0.492
## trt100 mg    -0.04974    0.07390 143.63000  -0.673   0.502
## avisitn15     0.07743    0.04677 132.54000   1.655   0.100
## avisitn29     0.06904    0.05650 134.52000   1.222   0.224
## avisitn43    -0.02690    0.07004 136.52000  -0.384   0.702
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Covariance estimate:
##      -7      15      29      43
## -7 0.2189 0.1789 0.1789 0.1789
## 15 0.1789 0.2189 0.1789 0.1789
## 29 0.1789 0.1789 0.2189 0.1789
## 43 0.1789 0.1789 0.1789 0.2189

```

```
aic_table <- tibble(
  Model = c("Unstructured", "Autoregressive", "Compound Symmetry"),
  AIC = c(AIC(m), AIC(m2), AIC(m3))
)
```

```
aic_table
```

```
## # A tibble: 3 x 2
##   Model      AIC
##   <chr>    <dbl>
## 1 Unstructured 106.
## 2 Autoregressive 105.
## 3 Compound Symmetry 104.
```

Conclusion

We see that the compound symmetry model gives the optimal fit according to AIC and will therefore be used in the primary analysis of ImPRESS. Note that we will use the corresponding restricted maximum likelihood estimates for the final model, and use mixed model with a random intercept for the modelling, but this is completely equivalent to the marginal model with compound symmetry covariance structure.